

Is the Stability Threshold n_{thresh} Universal?

Scale analysis of Matzke's 6.41-bit threshold

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Abstract

Douglas Matzke's hyperbit framework (IPI 2026) derives a stability threshold

$$n_{\text{thresh}} = \frac{2\pi}{\sqrt{2} \ln 2} = 6.4097 \text{ bits} \quad (.1)$$

and reads it as a universal number: $\text{Cl}(6)$ carries 6.000 bits and thus sits 0.41 bits below the collapse threshold – matter is stable because the smallest fermion algebra just barely fails to collapse. Doc. 326 raised the suspicion that this number hangs on the choice of a particular temperature. This document tests that numerically and settles it.

Main result [B]

With the system-dependent bit energy of FFGFT $E_{\text{bit}}(L) = \hbar c/L$ (Doc. 257) one obtains exactly

$$n_{\text{thresh}}(L) = \frac{L}{\sqrt{2} \ell_P} \quad (.2)$$

– **linear in the scale**, not an invariant number. Matzke's 6.41 corresponds exactly to the scale

$$L_{\text{Matzke}} = \frac{2\pi \ell_P}{\ln 2} = 9.0647 \ell_P. \quad (.3)$$

The "universal" bit value is the FFGFT bit energy, evaluated at about nine Planck lengths. The universality is a concealed choice of scale.

What remains *untouched* is the robust core of Matzke's argument: the collapse condition Compton = Schwarzschild yields $M_{\text{coll}} = m_P/\sqrt{2}$ purely geometrically – without any reference to bits, Landauer, or a temperature. That part stands independently **[B]**.

Verification script: 326_nthresh_skalenabhaengigkeit.py – 12 assertions, all passed.

.1 The question

Doc. 326 works out the central divergence between Matzke's framework and FFGFT: universal vs. system-dependent bit values. The suspicion formulated there was that Matzke's stability threshold "hangs on the postulate $T = T_P$ " and would itself be system-dependent under a system-dependent bit energy – classified as **[S]**.

This question is numerically decidable, and completely so: one recomputes Matzke's chain, extracts the scale-independent parts, replaces the bit value by the FFGFT bit energy, and sees whether a constant remains.

.2 Matzke's chain

The Landauer bridge

The temperature used is not the Planck temperature itself but the **KMS temperature** of the Planck scale:

$$T_{\text{KMS}} = \frac{T_P}{2\pi} = 2.2549 \times 10^{31} \text{ K}. \quad (.4)$$

The 2π comes from the KMS periodicity – the same one that generates the Hawking and Unruh temperatures in Doc. 325. Landauer at this temperature:

$$E_{\text{bit}} = k_B T_{\text{KMS}} \ln 2 \implies m_{\text{bit}} = \frac{m_P \ln 2}{2\pi} = 0.1103 m_P. \quad (.5)$$

Numerically reproduced (script [1], ratio 1.000000000). **[B]**

The collapse condition

Compton wavelength equal to Schwarzschild radius:

$$\frac{\hbar}{Mc} = \frac{2GM}{c^2} \implies M_{\text{coll}} = \sqrt{\frac{\hbar c}{2G}} = \frac{m_{\text{P}}}{\sqrt{2}} = 1.5390 \times 10^{-8} \text{ kg}. \quad (.6)$$

The scale-independent core [B]

$M_{\text{coll}} = m_{\text{P}}/\sqrt{2}$ contains **no bit value**. The collapse condition is purely geometric and identical under any bit interpretation. Numerical check: $n_{\text{thresh}} \cdot m_{\text{bit}} = M_{\text{coll}}$ to nine digits (script [2]).

The threshold

$$n_{\text{thresh}} = \frac{M_{\text{coll}}}{m_{\text{bit}}} = \frac{m_{\text{P}}/\sqrt{2}}{m_{\text{P}} \ln 2 / (2\pi)} = \frac{2\pi}{\sqrt{2} \ln 2} = 6.4097. \quad (.7)$$

The number is thus a *quotient* of a geometric numerator and a posited denominator. Everything about it that is not geometric sits in the bit value.

.3 The same computation with system-dependent bit energy

Instead of a universal bit value, FFGFT posits (Doc. 257, A271–A273):

$$E_{\text{bit}}(L) = \frac{\hbar c}{L} \implies m_{\text{bit}}(L) = \frac{\hbar}{cL}. \quad (.8)$$

Inserted into $n_{\text{thresh}} = M_{\text{coll}}/m_{\text{bit}}$:

$$n_{\text{thresh}}(L) = \frac{m_{\text{P}}}{\sqrt{2}} \cdot \frac{cL}{\hbar} = \frac{L}{\sqrt{2} \ell_{\text{P}}}, \quad (.9)$$

since $\ell_{\text{P}} = \hbar/(cm_{\text{P}})$. Numerically confirmed for $L \in \{1, 5, 10\} \ell_{\text{P}}$ to machine precision (script [3]). **[B]**

What follows

n_{thresh} is a **linear function of the scale**, not a constant. It has no distinguished value as long as no scale is distinguished. The question "how many bits until collapse?" is unanswerable without specifying the scale – it is underdetermined.

Matzke's implicit scale

Solving $m_{\text{bit}} = \hbar/(cL)$ for L and inserting Matzke's bit value:

$$L_{\text{Matzke}} = \frac{\hbar}{cm_{\text{bit}}} = \frac{2\pi \ell_{\text{P}}}{\ln 2} = 9.0647 \ell_{\text{P}} = 1.4651 \times 10^{-34} \text{ m}. \quad (.10)$$

And indeed Eq. (.2) returns $n_{\text{thresh}} = 6.4097$ at this scale – exactly Matzke's value (script [4]). **[B]**

Core finding

Matzke's universal bit value is the FFGFT bit energy $E_{\text{bit}} = \hbar c/L$, evaluated at $L = 2\pi\ell_P / \ln 2$. Both frameworks compute with the same quantity; the difference is that FFGFT carries the scale along as a free parameter, whereas Matzke fixes one particular scale and calls the result universal.

.4 Numerical range

n_{thresh} at the scales of the corpus (script [5]):

Scale	L [m]	n_{thresh}
$L_0 = \xi\ell_P$ (FFGFT minimal scale, Doc. 180)	2.155×10^{-39}	9.43×10^{-5}
ℓ_P (Planck length)	1.616×10^{-35}	0.7071
$2\pi\ell_P / \ln 2$ (Matzke's implicit scale)	1.465×10^{-34}	6.4097
1 fm (nuclear physics)	1.000×10^{-15}	4.38×10^{19}
$L_e = \hbar c/E_0$ (Doc. 257)	2.685×10^{-14}	1.17×10^{21}
1 nm (atomic physics)	1.000×10^{-9}	4.38×10^{25}

Range across these scales: a factor of 4.64×10^{29} . A quantity varying over 29 orders of magnitude is not a constant. **[B]**

Sensitivity to the temperature choice

Equivalently, the dependence can be expressed via the posited temperature (script [6]); n_{thresh} scales exactly inversely with it:

Posited temperature	m_{bit}/m_P	n_{thresh}
0.5 T_{KMS}	0.0552	12.82
0.9 T_{KMS}	0.0993	7.12
1.0 T_{KMS}	0.1103	6.41
1.1 T_{KMS}	0.1213	5.83
2.0 T_{KMS}	0.2206	3.20

A deviation of just 10 % in the posited temperature shifts n_{thresh} by $1 - 1/1.1 = 9.1$ % – and thereby moves Cl(6) from 0.41 bits below the threshold to 0.17 bits above it (at 1.1 T_{KMS} : $n_{\text{thresh}} = 5.83 < 6.000$). The stability claim flips.

.5 "It from bit" – with resolution depth

The finding of this document reads as criticism. It is, however, above all a *refinement*, and read that way it confirms Matzke's underlying intuition rather than refuting it.

The analogy to digital technology

In computing, a bit has no absolute meaning. “16 bits” says nothing until it is clear: 16 bits of *what*? A 16-bit colour value, a 16-bit audio sample and a 16-bit address space carry the same bit count and denote entirely different physical quantities. What a bit *means* is fixed only by the quantisation – the spacing between two distinguishable states. For audio that is the amplitude resolution, for images the colour depth, for instruments the size of the least significant bit.

$E_{\text{bit}} = \hbar c/L$ is a resolution depth

The FFGFT bit energy does not say “a bit weighs this much” but “at this scale a distinction costs this much”. $\hbar c$ is the exchange rate between distinction and energy; L states at which level the distinguishing happens. A bit at nuclear scale is a different physical object from a bit at atomic scale – although both carry exactly one yes/no decision.

Wheeler’s *it from bit* is thereby not refuted but sharpened: yes, physics can be built from distinguishability – but information is nothing absolute, it is always relative to a resolution. The missing parameter in the classical formulation is the resolution depth.

In this language Matzke’s choice of a fixed bit value is not wrong but a fixing of the *word width*: one fixes the quantisation and computes – a familiar and legitimate procedure from engineering practice. The only fallacy is to mistake the chosen word width for a constant of nature. What Matzke actually demonstrated – that the whole of black-hole physics can be formulated as pure bit bookkeeping – stands untouched and is the real result.

The floor of resolution

Unlike in digital technology, the resolution depth in FFGFT is bounded from below:

$$L \geq L_0 = \xi \cdot \ell_P = 2.155 \times 10^{-39} \text{ m} \quad (\text{Doc. 180 [K]}). \quad (.11)$$

L_0 is the counterpart of the smallest representable quantisation step. Below it there are no distinctions left – not because one could not measure precisely enough, but because there is nothing left to distinguish. The maximal resolution depth is thus finite, and precisely from this it follows in Doc. 327 that the spectral operator $\hat{F}$ is bounded; it is self-adjoint on the $\mathbb{Z}_3$ -invariant sector. There are no infinities in FFGFT. On why ℓ_P as a reference scale is not arbitrary – and where the justification of the factor ξ nevertheless remains open – see the following subsection.

Why ℓ_P is not an arbitrary scale

Here an obvious objection arises: if all scales are on an equal footing and only the resolution depth changes, why should there be a floor at all – and why precisely at ξ times the Planck length? Would that not be the very fallacy charged against Matzke: distinguishing a scale without justifying it?

The answer is in Section 3 of this document. ℓ_P is not one scale among many but precisely the one at which Compton wavelength and Schwarzschild radius coincide – that is, the collapse condition, which was shown there to be *purely geometric and free of any bit value* ($M_{\text{coll}} = m_P/\sqrt{2}$, with no reference to bits, Landauer, or temperature). The situation is therefore asymmetric:

Matzke's $L = 2\pi\ell_P/\ln 2$	FFGFT's ℓ_P
comes from a Landauer evaluation at a posited temperature	is the scale at which geometry itself tips over: below it no distinguishable object exists, because it collapses
has no independent justification – <i>chosen</i>	follows from ξ via $G = \xi^2/(4m_e)$ – <i>derived</i>

What matters is the direction of the derivation chain (Doc. 180):

$$\xi = \lambda_{\min}(\hat{F}_{D_4}) \implies G = \frac{\xi^2}{4m_e} \implies \ell_P = \sqrt{\frac{\hbar G}{c^3}} \implies L_0 = \xi \cdot \ell_P. \quad (.12)$$

ξ is a spectral value, determined independently of any length (Docs. 322, 327). In FFGFT ℓ_P is a *derived* quantity, not an input. The chain runs from the algebra to the length, not the reverse.

The floor follows from the Lagrangian [B] (R89/R90)

The last step of the chain is better founded than Doc. 180 presents it. There $L_0 = \xi^1 \cdot \ell_P$ is justified by the coupling term $g_T^l = \xi m_\ell$ carrying one power of ξ – an argument by analogy. The exponent, however, follows *necessarily* from the mass term: the time field Δm is a massive mediator with

$$m_T = \frac{M_{\text{base}}}{\xi} \implies \text{range} = \frac{\hbar}{m_T c} = \xi \cdot \lambda_C(M_{\text{base}}), \quad (.13)$$

hence $\text{range} \sim \xi^1$, because $m_T \sim \xi^{-1}$ and $\lambda_C \sim 1/m$. In the fundamental sector ($M_{\text{base}} = m_P$) the range is exactly $\xi \ell_P = L_0$ [B].

This also answers *why* a floor exists at all: because the mediator is massive. Below its Compton range the time field can no longer mediate – there “distance” loses its meaning. No postulate, but propagator physics.

The same rule accounts for the m_T values in the corpus only in part: Doc. 032 evaluates in the lepton sector ($M_{\text{base}} = m_e$); there the universal form gives $m_T = m_e/\xi \approx 3.83$ GeV. The 5.22 GeV of Doc. 032 (which itself calls it an *effective* torsion mass) follow only with additional factors. They result, moreover, only with $E_0 = 7500$ GeV from Doc. 032 ($E_0 \xi = 1$ GeV); with the corpus value $E_0 = 7.398$ MeV the same formula would give about 5 MeV. Universal form, sector-dependent value – structurally identical with $E_{\text{bit}} = \hbar c/L$.

What remains open [S]

The choice $M_{\text{base}} = m_P$ for the fundamental sector introduces no new parameter ($\lambda = 1$ in Planck units) but is not independently justified (R90).

The Schwarzschild check mentioned in Doc. 190 (Clarification 4) offers no support: an object with $M_0 = \xi m_{\text{P}}/2$ has Schwarzschild radius $\xi \ell_{\text{P}}$, but this is an *identity* – from $r_s = 2GM/c^2$ and $\ell_{\text{P}} = Gm_{\text{P}}/c^2$ it follows for any value of ξ and any exponent (R92). It tests nothing.

For the arguments of this document and for Doc. 327 this has no consequences: both require only that *some* finite floor exists, not its precise value. The number of rungs of the scale ladder is finite for any $L_0 = \xi^n \ell_{\text{P}}$ with finite n , so $\sum_n r_n < \infty$ and hence the boundedness of $\hat{F}$ hold independently of the exponent. What remains open is the *location* of the floor, not its existence.

The difference from Matzke's choice of scale thus persists and is sharper than the preceding analysis suggested: with Matzke the scale is chosen, and the choice carries the result. With FFGFT the reference scale ℓ_{P} is derived from ξ and the factor below it follows from the mediator mass; what remains open is only why the base mass of the fundamental sector is m_{P} .

The rules are scale-invariant, the values are not

The decisive point: that the *values* are scale-dependent does not mean the *rules* are. In FFGFT the same structure holds at every scale:

Rule	Form	scale-invariant
Time-mass duality	$\tilde{T} \cdot m = 1$	yes
Bit energy	$E_{\text{bit}} = \hbar c/L$	yes (form), no (value)
KMS read-off rule	$T = \hbar/(k_B \tau)$	yes
Information unit	$\Delta w = 1$ (winding quantum)	yes
$\mathbb{Z}_3$ triality selection	$T_R = 0$	yes
Fractal dimension	$D_f = 3 - \xi$	yes

The KMS rule, for instance, tempers Unruh, Gibbons–Hawking and black holes with *one* formula across 22 orders of magnitude in τ (Docs. 313, 325). The triality selection $T_R = 0$ acts in hadron confinement just as it does at the horizon of a black hole (Docs. 321, 325). The winding quantum $\Delta w = 1$ is the same topological quantity at every scale – only its energy is scale-specific.

The actual finding

Scale-invariant *rules* with scale-dependent *values* are no contradiction but the normal case in any field theory – consider running couplings: $\alpha_s(\mu)$ changes its value with the scale, while the Callan–Symanzik equation describing that running holds scale-independently.

$E_{\text{bit}}(L) = \hbar c/L$ behaves in exactly this way: the *formula* is universal, the *value* is not. Whoever takes the value to be universal mistakes the rule for one of its evaluations.

With this, “it from bit” can be stated precisely in FFGFT: the world is built from distinctions, and the rules by which distinctions compose into physics are the same at all scales. What changes with the scale is only the price of a distinction – the

resolution depth. Matzke's 6.41 is then no error but the right number for the word width $L = 2\pi\ell_P / \ln 2$.

.6 Placement

What falls

The reading of $n_{\text{thresh}} = 6.41$ as a *fundamental number* of the universe. It is the quotient of a geometric quantity and a posited bit value; the postulate is a choice of scale, and across scales the result varies by 29 orders of magnitude. The statement "Cl(6) lies 0.41 bits below the threshold" holds only at $L = 9.06 \ell_P$ and does not survive beyond it. **[X]**

What stands

Considerably more than the criticism might suggest:

- $M_{\text{coll}} = m_P / \sqrt{2}$ from Compton = Schwarzschild – purely geometric, independent of any bit interpretation **[B]**.
- The Schwarzschild radius without field equations: Matzke's Novel Result #1 is untouched by this analysis **[B]**.
- Cl(6) as a fermion generation (Furey), the three falling out, the algebraic structure as a whole **[B]**.
- The Hawking mechanism as an algebraic picture – the structural correspondences to FFGFT are exact (Doc. 326).

Consequence for FFGFT

The analysis is not merely a refutation; it supports the FFGFT position of Doc. 257 and A271–A273 from the outside. If a framework that explicitly posits a universal bit value can be reconstructed as the evaluation of the scale-dependent formula $E_{\text{bit}} = \hbar c / L$ at one particular scale, then scale dependence is not the special case but the general form.

This fits with Doc. 325: there the Hawking temperature is derivable without a universal bit value ($k_B T_H = E_{\text{bit}}(4\pi r_s)$ exactly), and the numerical check in Doc. 326 gave $T_H(\text{Matzke}) / T_H(\text{KMS}) = 1.000000$. Both findings say the same thing: the universal bit value is a reparametrization, not an independent physical input.

Worth noting is the origin of the 2π in Matzke's bit value: it comes from the KMS periodicity, i.e. from exactly the relation that generates the Hawking temperature in Doc. 325 *without* a bit value.

.7 Status summary

Statement	Status
$m_{\text{bit}} = m_{\text{P}} \ln 2 / (2\pi)$ from Landauer at $T_{\text{P}} / (2\pi)$ reproduced	[B]
$M_{\text{coll}} = m_{\text{P}} / \sqrt{2}$ purely geometric, bit-value free	[B]
$n_{\text{thresh}}(L) = L / (\sqrt{2} \ell_{\text{P}})$ – linear in the scale	[B]
Matzke's scale: $L = 2\pi \ell_{\text{P}} / \ln 2 = 9.0647 \ell_{\text{P}}$	[B]
Range across corpus scales: factor 4.6×10^{29}	[B]
$n_{\text{thresh}} \propto 1/T$ – a 10 % temperature error flips the claim	[B]
$n_{\text{thresh}} = 6.41$ as a universal number	[X]
Compton = Schwarzschild, $\text{Cl}(6)$ structure, r_s without GR	[B] (untouched)

Relation to the corpus

Building block	Doc.
Bit energy $E_{\text{bit}} = \hbar c / L$, information unit $\Delta w = 1$	257
Landauer as carrier operation, not abstract information	A271–A273
Elementary cells, bit capacity, partition invariance	302
$\ell_{\text{P}}, m_{\text{P}}, L_0 = \xi \ell_{\text{P}}$	180
KMS relation, $k_B T_H = E_{\text{bit}}(4\pi r_s)$	325
Matzke comparison, bit-value divergence	326

References

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